**3** Solution **A** and solid **B** were analysed. Solution **A** was aqueous copper(II) bromide.
Tests were done on each substance.

Complete the expected observations.

**tests on solution A**

Solution **A** was divided into three approximately equal portions in three test-tubes.

**(a)** The end of a piece of wire was dipped into the first portion of solution **A**. The end of the wire was then placed at the edge of a roaring Bunsen burner flame.

observations .................................................................................................................. [1]

**(b)** To the second portion of solution **A** aqueous ammonia was added dropwise until in excess.

observations ......................................................................................................................

..........................................................................................................................................

.................................................................................................................................. [3]

**(c)** To the third portion of solution **A** about 1 cm depth of dilute nitric acid followed by a few drops of aqueous silver nitrate were added.

observations ......................................................................................................................

.................................................................................................................................. [1]

**tests on solid B**

| tests | observations |
|---|---|
| Solid **B** was added to 15 $cm^3$ of water in a boiling tube. A bung was placed in the boiling tube and it was shaken to dissolve solid **B** and form solution **B**. Solution **B** was divided into three approximately equal portions in three test-tubes.<br><br>**test 1**<br><br>The first portion of solution **B** was tested using universal indicator paper. | the universal indicator paper turned blue |
| **test 2**<br><br>To the second portion of solution **B** aqueous sodium hydroxide was added dropwise and then in excess. | a white precipitate formed which remained when excess aqueous sodium hydroxide was added |
| **test 3**<br><br>To the third portion of solution **B** aqueous ammonia was added dropwise and then in excess. | the solution remained colourless |

**(d)** Deduce the pH of solution **B**.

pH = .............................. [1]

**(e)** Identify solid **B**.

..............................................................................................................................

.............................................................................................................................. [2]

[Total: 8]

**3** Solid **E** and solution **F** were analysed. Solid **E** was ammonium sulfate.
Tests were done on each substance.

**tests on solid E**

Complete the expected observations.

Solid **E** was dissolved in water to form solution **E**. Solution **E** was divided into three approximately equal portions in one boiling tube and two test-tubes.

**(a)** Aqueous sodium hydroxide was added to the first portion of solution **E** in a boiling tube. The mixture formed was warmed. Any gas produced was tested.

observations ..............................................................................................................

..............................................................................................................

identity of gas ..............................................................................................................
[2]

**(b)** To the second portion of solution **E**, about 1 cm depth of dilute nitric acid followed by a few drops of aqueous silver nitrate were added.

observations .............................................................................................................. [1]

**(c)** To the third portion of solution **E**, about 1 cm depth of dilute nitric acid followed by a few drops of aqueous barium nitrate were added.

observations .............................................................................................................. [1]

**tests on solution F**

| tests | observations |
|---|---|
| Solution **F** was divided into two equal portions in two test-tubes.<br><br>**test 1**<br><br>A strip of universal indicator paper was placed in the first portion of solution **F**. | the universal indicator paper turned orange |
| **test 2**<br><br>The second portion of solution **F** was added to solid sodium carbonate in a boiling tube.<br>Any gas made was tested. | effervescence and the solid disappeared<br><br>limewater turned milky |

**(d)** Deduce the pH of solution **F**.

.............................................................................................................................................. [1]

**(e)** Identify the positive ion in solution **F**.

.............................................................................................................................................. [1]

[Total: 6]

**3** Solid **M** and solid **N** were analysed. Solid **M** was iron(III) nitrate.
Tests were done on each substance.

**tests on solid M**

Complete the expected observations.

Solid **M** was dissolved in water to form solution **M**. Solution **M** was divided into two approximately equal portions in two test-tubes.

**(a)** To the first portion of solution **M**, aqueous sodium hydroxide was added gradually until in excess. The product was kept for **(b)**.

observations ....................................................................................................................

.................................................................................................................................. [2]

**(b)** **(i)** The product from **(a)** was transferred to a boiling tube. A piece of aluminium foil was added and the mixture warmed gently. Any gas produced was tested.

observations ....................................................................................................................

.................................................................................................................................. [1]

**(ii)** Identify the gas made in **(i)**.

.................................................................................................................................. [1]

**(c)** To the second portion of solution **M**, about 1 cm depth of dilute nitric acid followed by a few drops of aqueous barium nitrate were added.

observations .................................................................................................................. [1]

**tests on solid N**

| tests | observations |
|---|---|
| **test 1**<br><br>A flame test was carried out on solid **N**. | the flame became red |
| Solid **N** was dissolved in water to form solution **N**. Solution **N** was divided equally into one test-tube and one boiling tube.<br><br>**test 2**<br><br>About 1 cm depth of dilute nitric acid followed by a few drops of aqueous silver nitrate were added to the first portion of solution **N** in a test-tube. | no visible change |
| **test 3**<br><br>About 2 cm depth of dilute hydrochloric acid was added to the second portion of solution **N**. The mixture was warmed and any gas produced was tested. | acidified aqueous potassium manganate(VII) changed from purple to colourless |

**(d)** Identify the gas produced in **test 3**.

.............................................................................................................................. [1]

**(e)** Identify solid **N**.

..............................................................................................................................

.............................................................................................................................. [2]

[Total: 8]

**3** Solid **S** and solution **Y** were analysed. Solid **S** was anhydrous copper(II) sulfate.
Tests were done on each substance.

**tests on solid S**

Complete the expected observations.

**(a)** A flame test was carried out on solid **S**.

observations .............................................................................................................. [1]

The remaining solid **S** was dissolved in about 10 $cm^3$ of distilled water to form solution **T**. Solution **T** was divided into two approximately equal portions in two test-tubes.

**(b)** State the colour change that occurred when water was added to solid **S** to form solution **T**.

from solid **S** ................................................ to solution **T** ................................................ [1]

**(c)** To the first portion of solution **T**, about 1 cm depth of dilute nitric acid followed by a few drops of aqueous barium nitrate were added.

observations .............................................................................................................. [1]

**(d)** To the second portion of solution **T**, aqueous ammonia was added dropwise and then in excess.

observations ..............................................................................................................

..............................................................................................................

.............................................................................................................. [3]

**tests on solution Y**

| tests | observations |
|---|---|
| **test 1**<br><br>A flame test was carried out on solution **Y**. | the flame became lilac |
| Solution **Y** was divided into three approximately equal portions in one boiling tube and two test-tubes.<br><br>**test 2**<br><br>Dilute hydrochloric acid was added to the portion of solution **Y** in a boiling tube. The mixture was warmed. A strip of filter paper soaked in acidified aqueous potassium manganate(VII) was held at the mouth of the boiling tube. | the acidified aqueous potassium manganate(VII) remained purple |
| **test 3**<br><br>About 1 cm depth of dilute nitric acid followed by a few drops of aqueous silver nitrate were added to the second portion of solution **Y**. | yellow precipitate |
| **test 4**<br><br>Aqueous ammonia was added dropwise and then in excess to the third portion of solution **Y**. | a white precipitate formed which dissolved in excess to give a colourless solution |

**(e)** Name the gas tested for in **test 2**.

.............................................................................................................................. [1]

**(f)** Identify the **three** ions in solution **Y**.

..............................................................................................................................

.............................................................................................................................. [3]

[Total: 10]

**3** Two solids, solid **C** and solid **D**, were analysed.
Tests were done on each solid.

**tests on solid C**

Tests were carried out and the following observations were made.

| tests | observations |
|---|---|
| **test 1**<br><br>A flame test was carried out on solid **C**. | a red flame was seen |
| Solid **C** was dissolved in distilled water to produce solution **C**.<br><br>**test 2**<br><br>About 5 $cm^3$ of aqueous sodium hydroxide was added to solution **C**. | no change |
| **test 3**<br><br>A piece of aluminium foil was added to the mixture formed in **test 2**.The mixture was warmed gently and any gas produced was tested. | effervescence was seen;<br>damp red litmus paper turned blue |

**(a)** Name the gas that turned the damp red litmus paper blue in **test 3**.

.................................................................................................................................... [1]

**(b)** Identify solid **C**.

....................................................................................................................................

.................................................................................................................................... [2]

**tests on solid D**

Solid **D** was aluminium sulfate.

Complete the expected observations.

Solid **D** was dissolved in water to form solution **D**. Solution **D** was divided into four approximately equal portions in four test-tubes.

**(c)** Aqueous sodium hydroxide was added dropwise and then in excess to the first portion of solution **D**.

observations ..............................................................................................................

.............................................................................................................................. [2]

**(d)** Aqueous ammonia was added dropwise and then in excess to the second portion of solution **D**.

observations ..............................................................................................................

.............................................................................................................................. [2]

**(e)** About 1 $cm^3$ of dilute nitric acid and a few drops of aqueous silver nitrate were added to the third portion of solution **D**.

observations ........................................................................................................ [1]

**(f)** About 1 $cm^3$ of dilute nitric acid and a few drops of aqueous barium nitrate were added to the fourth portion of solution **D**.

observations ........................................................................................................ [1]

[Total: 9]

**3** Solid **E** and solution **F** were analysed.
Tests were done on each substance.

**tests on solid E**

| tests | observations |
|---|---|
| **test 1**<br><br>About half of solid **E** was placed in a test-tube and heated gently. | steam was given off;<br>condensation appeared near the mouth of the test-tube |
| The remaining solid **E** was dissolved in distilled water to produce solution **E**. The solution was divided into four equal portions in three test-tubes and a boiling tube.<br><br>**test 2**<br><br>About 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous silver nitrate were added to the first portion of solution **E**. | no visible change |
| **test 3**<br><br>About 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous barium nitrate were added to the second portion of solution **E**. | white precipitate |
| **test 4**<br><br>Excess aqueous ammonia was added to the third portion of solution **E**. | white precipitate |
| **test 5**<br><br>Aqueous sodium hydroxide was added dropwise and then in excess to the fourth portion of solution **E** in the boiling tube. | white precipitate which dissolved in excess to form a colourless solution |
| **test 6**<br><br>The product from **test 5** was warmed gently and any gas given off was tested with damp red litmus paper. | the red litmus paper turned blue |

**(a)** State the conclusion that can be made from the observations in **test 1**.

..............................................................................................................................................

.............................................................................................................................................. [1]

**(b)** State the conclusion that can be made from the observation in **test 2**.

..............................................................................................................................................

.............................................................................................................................................. [1]

**(c)** Identify the **three** ions in solid **E**.

..........................................................................................................................................

.................................................................................................................................... [3]

**tests on solution F**

Solution **F** was aqueous sodium hydroxide.

Complete the expected observations.

**(d)** A flame test was carried out on solution **F**.

observations .................................................................................................... [1]

**(e)** The remaining solution **F** was divided into two approximately equal portions in two test-tubes.

**(i)** To the first portion of solution **F** a few drops of universal indicator solution were added.

observations .............................................................................................. [1]

**(ii)** To the second portion of solution **F** approximately 2 $cm^3$ of aqueous copper(II) sulfate was added.

observations .............................................................................................. [1]

[Total: 8]

**3** Solution **G** and solid **H** were analysed.

**tests on solution G**

| tests | observations |
|---|---|
| Solution **G** was divided into three equal portions in three test-tubes.<br><br>**test 1**<br><br>Sodium hydroxide was added dropwise and then in excess to the first portion of solution **G**. | white precipitate which did not dissolve in excess |
| **test 2**<br><br>About 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous silver nitrate were added to the second portion of solution **G**. | yellow precipitate |
| **test 3**<br><br>About 10 $cm^3$ of aqueous hydrogen peroxide was added to the third portion of solution **G**. The gas produced was tested. | the mixture became brown and bubbled; the gas relit a glowing splint |

**(a)** Identify the gas produced in **test 3**.

.......................................................................................................................... [1]

**(b)** Use the results of **test 1** and **test 2** to identify solution **G**.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**tests on solid H**

Solid **H** was hydrated copper(II) sulfate.

Complete the expected observations.

**(c)** About half of solid **H** was placed in a boiling tube and heated using a Bunsen burner.

observations .............................................................................................................

............................................................................................................................. [2]

**(d)** A flame test was carried out on solid **H**.

observations ........................................................................................................ [1]

The remaining solid **H** was placed in a boiling tube. About 10 $cm^3$ of distilled water was added to the boiling tube. The tube was shaken to dissolve solid **H** and form solution **H**.

Solution **H** was divided into two approximately equal portions in two test-tubes.

**(e)** Aqueous ammonia was added dropwise and then in excess to the first portion of solution **H**.

observations .............................................................................................................

.............................................................................................................................

............................................................................................................................. [3]

**(f)** Approximately 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous barium nitrate were added to the second portion of solution **H**.

observations ........................................................................................................ [1]

[Total: 10]

**3** Solid **I** and solid **J** were analysed. Solid **I** was chromium(III) chloride.

**tests on solid I**

Complete the expected observations.

Solid **I** was placed in a boiling tube and about 10 $cm^3$ of distilled water was added to the boiling tube. The mixture was shaken to dissolve solid **I** and form solution **I**. Solution **I** was divided into four portions in four test-tubes.

**(a)** Aqueous sodium hydroxide was added dropwise and then in excess to the first portion of solution **I**.

observations ..............................................................................................................

..............................................................................................................................

.............................................................................................................................. [2]

**(b)** Aqueous ammonia was added dropwise and then in excess to the second portion of solution **I**.

observations ..............................................................................................................

..............................................................................................................................

.............................................................................................................................. [2]

**(c)** About 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous silver nitrate were added to the third portion of solution **I**.

observations .............................................................................................................. [1]

**(d)** About 1 $cm^3$ of dilute nitric acid followed by a few drops of aqueous barium nitrate were added to the fourth portion of solution **I**.

observations .............................................................................................................. [1]

**tests on solid J**

| tests | observations |
|---|---|
| **test 1**<br><br>A flame test was carried out on solid **J**. | lilac flame |
| The remaining solid **J** was placed in a boiling tube and about 10 $cm^3$ of distilled water was added to the boiling tube. The mixture was shaken to dissolve solid **J** and form solution **J**.<br><br>**test 2**<br><br>About 5 $cm^3$ of dilute nitric acid was added to solution **J**.<br><br>Any gas produced was tested. | <br><br><br><br><br>effervescence<br><br>the gas turned limewater milky |
| **test 3**<br><br>A few drops of aqueous silver nitrate were added to the mixture formed in **test 2**. | no visible change |

**(e)** Identify the gas formed in **test 2**.

.............................................................................................................................................. [1]

**(f)** Identify solid **J**.

..............................................................................................................................................

.............................................................................................................................................. [2]

[Total: 9]

## Notes for use in qualitative analysis

### Tests for anions

| anion | test | test result |
|---|---|---|
| carbonate, $CO_3^{2-}$ | add dilute acid, then test for carbon dioxide gas | effervescence, carbon dioxide produced |
| chloride, $Cl^-$ [in solution] | acidify with dilute nitric acid, then add aqueous silver nitrate | white ppt. |
| bromide, $Br^-$ [in solution] | acidify with dilute nitric acid, then add aqueous silver nitrate | cream ppt. |
| iodide, $I^-$ [in solution] | acidify with dilute nitric acid, then add aqueous silver nitrate | yellow ppt. |
| nitrate, $NO_3^-$ [in solution] | add aqueous sodium hydroxide, then aluminium foil; warm carefully | ammonia produced |
| sulfate, $SO_4^{2-}$ [in solution] | acidify with dilute nitric acid, then add aqueous barium nitrate | white ppt. |
| sulfite, $SO_3^{2-}$ | add a small volume of acidified aqueous potassium manganate(VII) | the acidified aqueous potassium manganate(VII) changes from purple to colourless |

### Tests for aqueous cations

| cation | effect of aqueous sodium hydroxide | effect of aqueous ammonia |
|---|---|---|
| aluminium, $Al^{3+}$ | white ppt., soluble in excess, giving a colourless solution | white ppt., insoluble in excess |
| ammonium, $NH_4^+$ | ammonia produced on warming | – |
| calcium, $Ca^{2+}$ | white ppt., insoluble in excess | no ppt. or very slight white ppt. |
| chromium(III), $Cr^{3+}$ | green ppt., soluble in excess | grey-green ppt., insoluble in excess |
| copper(II), $Cu^{2+}$ | light blue ppt., insoluble in excess | light blue ppt., soluble in excess, giving a dark blue solution |
| iron(II), $Fe^{2+}$ | green ppt., insoluble in excess, ppt. turns brown near surface on standing | green ppt., insoluble in excess, ppt. turns brown near surface on standing |
| iron(III), $Fe^{3+}$ | red-brown ppt., insoluble in excess | red-brown ppt., insoluble in excess |
| zinc, $Zn^{2+}$ | white ppt., soluble in excess, giving a colourless solution | white ppt., soluble in excess, giving a colourless solution |

**Tests for gases**

| gas | test and test result |
|---|---|
| ammonia, $NH_3$ | turns damp red litmus paper blue |
| carbon dioxide, $CO_2$ | turns limewater milky |
| chlorine, $Cl_2$ | bleaches damp litmus paper |
| hydrogen, $H_2$ | 'pops' with a lighted splint |
| oxygen, $O_2$ | relights a glowing splint |
| sulfur dioxide, $SO_2$ | turns acidified aqueous potassium manganate(VII) from purple to colourless |

**Flame tests for metal ions**

| metal ion | flame colour |
|---|---|
| lithium, $Li^+$ | red |
| sodium, $Na^+$ | yellow |
| potassium, $K^+$ | lilac |
| calcium, $Ca^{2+}$ | orange-red |
| barium, $Ba^{2+}$ | light green |
| copper(II), $Cu^{2+}$ | blue-green |